
UNIT 8 DECISION MAKING

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8.0 INTRODUCTION

Artificial Intelligence (AI) has become deeply embedded in nearly every aspect of contemporary life. Its applications span a wide spectrum of domains, including smartphone voice assistants, entertainment systems, financial services, and healthcare technologies. Across these varied fields, decision-making constitutes a central and indispensable component of AI systems. One foundational concept underpinning intelligent decision-making is utility theory, which provides a structured approach for evaluating and selecting among alternative courses of action.

To fully comprehend how AI systems make informed choices, it is essential to understand the mathematical framework that governs decision-making processes, particularly in situations where multiple options are available. Within this framework, the concept of an autonomous rational agent—commonly referred to simply as an agent—is fundamental. An agent can be defined as an entity, often implemented as a software program or bot, that perceives its environment, evaluates possible actions, and selects those that best achieve its specified objectives. The choice of action is determined by the agent's perception of its current situation and its goal-directed reasoning aimed at maximizing expected utility.

8.1 OBJECTIVES

After going through this unit, you should be able to:

- Understand Decision theory and Utility theory
- Solve sequential decision problem
- Understand game theory and its application

8.2 BASICS OF UTILITY THEORY

One of the fundamental attributes associated with human intelligence is the ability to make sound and rational decisions. In contrast to human decision-making, which is often influenced by intuition, experience, and subjective judgment, decision-making in Artificial Intelligence (AI) systems is primarily grounded in mathematical models and algorithmic processes. When decisions must be made in alignment with clearly defined objectives, a formal framework is necessary to systematically optimize outcomes. Utility theory in AI offers such a framework by enabling the incorporation of critical factors such as uncertainty, risk preferences, and perceptual information into the decision-making process, thereby guiding agents toward choices that maximize expected benefit.

In every decision-making process, there is a degree of uncertainty involved. Sometimes, the level of uncertainty is higher or lower, which depends on the number of contributing factors. The utility theory provides a mathematical framework for incorporating these uncertainty factors, which help an autonomous AI agent make decisions. Mathematical models are used to optimize the decision-making process by assigning numerical values or functions to evaluate the subjective values or preferences. The concept of utility theory is extensively applied not only in various domains of Artificial Intelligence—such as robotics, recommender systems, and game-theory models—but also across disciplines including economics, finance, and healthcare, where decision-making under uncertainty plays a crucial role.

More fundamentally, the utility theory facilitates the decision-making in AI to achieve a specific goal called utility. The utility or perceived goal varies from person to person, for example, in robotics, the utility could be assumed as the cost of navigation by one person with a reasonable degree of risk involved. Another individual with a higher risk appetite may minimize navigation costs by opting for routes associated with greater risk probabilities. Another way to understand utility is through environmental probabilities, where the end result depends on these probabilities. For example, in self-driving cars, the utility function is dependent on various components: traffic conditions, safety, fuel consumption, etc. Utility values are estimated using different functions for each condition, and the appropriate action (decision) is then determined.

8.2.1 Utility Function

The utility function is a measure for quantifying preferences in an environment. It is generally computed by using a mathematical function that returns a numerical value. The utility function plays a crucial role in representing human satisfaction and should therefore be chosen carefully according to the specific problem domain. The utility function depends on the desired goal and available choices. It maps the desired outcomes to their corresponding utility values. The utility is denoted by U .

The utility of a state S is represented by the mathematical function $U(S)$, where S denotes a particular state of the environment. A state may be understood as the current situation in which the autonomous agent finds itself. When the

agent acts, the state of the environment changes accordingly. The selection of an action is guided by the utility associated with the resulting states, to maximize overall utility. In subsequent discussions, we will examine how complex situations can be simplified through the decomposition of states, thereby reducing the complexity of decision-making.

Utility Principle: A real-valued utility U that operates on the states A and B ,
 if $U(A) > U(B)$ then agents prefer A over B .
 if $U(A) = U(B)$ then agent is indifferent of A and B .

Example 1: To understand more about the mathematical function for utility, let's take a simple example of playing a gamble. Suppose a gambler has to choose between the following two:

- i) First move will give a guaranteed prize of Rs. 1000.
- ii) Second move, provides 50% chance of winning Rs. 1500 and 50% probability of losing.

Which option may a gambler choose?

His choice is influenced by many factors, such as the degree of risk-taking ability or financial condition. The utility function maps these influencing factors by means of a suitable mathematical function, which eventually helps in making a final decision. If we associate the utility function simply with some monetary value, like the utility of the first move is Rs. 1000 and the utility of the second move is Rs. 1500, then the second move is preferable.

Utility Axioms

The utility theory specifies certain constraints on rational preferences. These constraints are defined as the axioms of utility theory. These axioms serve as the foundation for understanding the usage of utility functions to model decision-making under uncertain circumstances. The following are the six utility axioms:

A. Completeness

Given any two states, a rational agent prefers one over the other based on their utility values. In other words, if $U(A) > U(B)$, then outcome(A) is preferred to outcome(B).

B. Transitivity

Given any three states, if a rational agent preferred A to B , and B is preferred to C , then A should be preferred to C . This axiom ensures that the preferences modeled by the utility function are consistent and do not lead to contradictions.

$$(U(A) > U(B)) \quad \cap \quad (U(B) > U(C)) \quad \rightarrow \quad (U(A) > U(C))$$

C. Continuity

Small variations in the probabilities of outcomes should lead to correspondingly small variations in expected utility. For instance, if state B is preferred over state C but less preferred than state A , then there exists some probability p such that a rational agent is indifferent between receiving B with certainty and participating in a lottery that yields A with probability p and C with probability $1-p$. This axiom ensures continuity in preferences, implying that the utility function is smooth and well-behaved, and that minor changes in probabilities do not produce abrupt or inconsistent shifts in decision-making.

D. Substitutability

If two states, A and B , are equally preferred, the agent's choice is indifferent between two more complex states that are the same except that B is substituted for A in one of them. This axiom allows for substituting equally preferred outcomes without affecting the decision-making process.

If B is substituted for A , it implies that, with whatever probability p , A is selected, the probability of choosing B is also p .

E. Monotonicity

If the probability of a particular outcome increases, its expected utility should increase accordingly. This axiom reflects the principle of monotonicity, ensuring that a higher likelihood of a preferred outcome enhances its overall perceived utility. For example, consider two lotteries that yield the same outcomes, A and B . If an agent prefers A over B , then the agent will prefer the lottery that assigns a higher probability to outcome A compared to the probability assigned to B .

F. Decomposability

The utility function should be able to represent preferences over multiple attributes or features of an outcome in a decomposable manner. Compound lotteries can be reduced to simpler lotteries using the probability theory laws. This allows for modeling complex decision problems with multiple dimensions or criteria.

The above-mentioned axioms of utility theory do not say about utility but help in concluding preferences. Preference is assumed to be a basic property of an agent. The utility function derived from these axioms forms the basis of the Utility Principle and the Maximum Expected Utility Principle.

8.2.2 Maximum Expected Utility Principle

Expected utility is one of the core concepts in the decision-making process to solve problems in AI and related fields. In the real world, the sequence of actions that are taken is mostly nondeterministic in nature. Deterministic

actions have definite outcomes, whereas nondeterministic actions are chosen based on a probabilistic assessment of the situation. Let's say a nondeterministic action A has possible outcomes denoted by $Outcomes_i(A)$, where i is the range of different outcomes. The agents assign probabilities $P(Outcomes_i(A)|E)$ to different conditions based on available evidence E . The expected utility denoted by $EU(A|E)$ is calculated as the multiplication of the assigned probability by the utility U , shown as below:

$$EU(A|E) = \sum_i P(Outcomes_i(A)|E) * U(Outcomes_i(A))$$

The Maximum Expected Utility (MEU) Principle states that a rational agent selects the action that maximizes its expected utility. To achieve a particular goal, the agent evaluates the possible sequences of actions and chooses the one that yields the highest expected utility. However, in complex environments, exhaustively enumerating all possible action sequences is computationally infeasible, especially when the decision horizon is long or the state space is large.

Let's take the same Example 1. The gambler selects a move based on the principle of expected utility. Suppose the first move yields a single certain outcome, denoted as $M1$. Since this outcome is guaranteed, its probability is $P(M1)=1$, and its associated utility is $U(M1)=$ Rs. 1000. Therefore, the expected utility of the first move is:

$$EU(\text{First}) = P(M1) * U(M1) = 1000.$$

Similarly, If the gambler choose second move, it has two outcomes $M2$ and $M3$, with probability of 50% each. The utility associated with the two outcomes are $U(M2) =$ Rs.1500 and $U(M3) = 0$. The expected utility is computed as:

$$\begin{aligned} EU(\text{Second}) &= P(M2) * U(M2) + P(M3) * U(M3) \\ &= 0.5 * \text{Rs.1500} + 0.5 * 0 \\ &= 750. \end{aligned}$$

According to the Maximum Expected Utility principle, the first move is preferred when it yields a higher expected utility. However, if the winning amount in the second move increases from Rs.1500 to Rs.4000, its expected utility rises to Rs.2000, making the second move preferable. Importantly, utility should not be equated directly with monetary value. In autonomous decision-making, utility typically reflects subjective valuation, and for most individuals it is concave over gains, indicating risk aversion.

Risk attitudes may vary depending on context. For instance, an individual already in debt may avoid a gamble offering a 50% chance of winning Rs.1500 and a 50% chance of losing Rs.1000, whereas someone with Rs.10,000 in savings may accept it. The amount an agent is willing to accept instead of participating in a

gamble is known as the certainty equivalent. Empirical studies suggest that many individuals would prefer a guaranteed Rs.1000 over a gamble that pays Rs.1500 with probability 0.5 and zero otherwise. The difference between the expected monetary value and the certainty equivalent is termed the risk premium, which underlies the insurance industry: risk-averse agents are willing to pay a premium lower than the expected loss to avoid uncertainty, such as purchasing insurance against potential damage to a house.

Check Your Progress 1

1. List the advantages and disadvantages of the Utility Function.

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2. Let us assume a person has an initial wealth of Rs.100000 with two options:

- (i) Invest (I): with probability 0.6 you move to wealth Rs.150000 and with probability 0.4 you move to Rs.60000.
- (ii) Keep (K): stay at wealth Rs.100000 with certainty.

The utility is taken of final wealth. Which action should you choose (based on maximum utility function)?.

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3. How maximum utility principal helps in making decision?

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8.3 DECISION THEORY

Decision-making in Artificial Intelligence (AI) involves selecting the most appropriate action from a set of alternatives to achieve a specified objective. These choices are guided by formal models that evaluate possible states, anticipate outcomes, and optimize criteria such as performance, utility, or reward.

While everyday decisions may rely on intuition, complex managerial, engineering, or computational problems require systematic and quantitative analysis. A unified decision-theoretic framework enables an agent to select a course of action under varying degrees of uncertainty and information availability. Such frameworks also address sequential decision problems, in

which current choices influence future states, opportunities, and risks, thereby necessitating dynamic analytical tools such as decision trees and related optimization methods.

8.3.1 Decision Theory: Foundations and Environments

Decision theory provides an analytical framework for choosing the best action when faced with uncertainty about the future environment. It forces the decision-maker to clearly identify all available alternatives and the potential outcomes conditioned on external events.

Core Components of a Decision Problem

A standard decision-theoretic problem consists of the following elemental components:

- The Decision Maker (DM): The individual or entity responsible for making the choice.
- Alternative Courses of Action (Acts): The set of mutually exclusive options available to the DM. We denote these as $A_1, A_2, \dots, A_n$.
- States of Nature (Events): Factors outside the DM's control that influence the outcome of the decision. These must be mutually exclusive and collectively exhaustive. We denote these as $S_1, S_2, \dots, S_m$.
- Payoffs (Outcomes): The numerical value (profit, cost, utility) resulting from a specific combination of an action chosen and a state of nature that occurs.

The relationship between actions, states of nature, and payoffs is best represented schematically by the payoff matrix shown in Table 1.

Table 1: The general payoff matrix structure

Alternative Courses of Action	States of Nature				
	State S_1	State S_2	State S_3	...	State S_m
Action A_1	Payoff 11	Payoff 12	Payoff 13	...	Payoff 1m
Action A_2	Payoff 21	Payoff 22	Payoff 23	...	Payoff 2m
Action A_3	Payoff 31	Payoff 32	Payoff 33	...	Payoff 3m
⋮	⋮	⋮	⋮	⋮	⋮
Action A_n	Payoff n1	Payoff n2	Payoff n3	...	Payoff nm

8.3.2 Enumeration of Decision Environments and Criteria

Decision theory categorizes problems based on the level of knowledge the DM possesses regarding the states of nature.

A. Decision Making Under Certainty

The DM knows exactly which state of nature will occur. The decision problem is trivial: simply select the action that yields the highest payoff for that specific state.

B. Decision Making Under Uncertainty (No Probabilities)

The DM knows the possible states of nature but has no information (or is unwilling to assign estimates) regarding the probability of these states occurring. In this environment, decisions are based on the DM's psychological attitude toward risk.

Common criteria used in this environment include:

i) The Maximin Criterion (Pessimistic): The DM assumes that the worst possible state will occur for whatever action they choose.

Procedure: Identify the minimum payoff for each action. Choose the action with the largest (maximum) of these minimums. This "maximizes the minimum gain."

ii) The Maximax Criterion (Optimistic): The DM assumes that the best possible state will occur.

Procedure: Identify the maximum payoff for each action. Choose the action with the largest of these maximums.

C. Decision Making Under Risk (Known Probabilities)

This is the most common business scenario. The DM does not know which state will occur, but they have probability estimates for each state ($P(S_1)$, $P(S_2)$, etc.). The sum of these probabilities must equal to 1.0. The primary criterion used here is the Expected Monetary Value (EMV). The EMV is the weighted average payoff for an action, where the weights are the probabilities of the states of nature.

Procedure for EMV computation: For each action, multiply every possible payoff by the probability of that state occurring and sum the results. Select the action with the highest EMV.

$$EMV(A_i) = \sum [Payoff(A_i, S_j) * P(S_j)]$$

Where Payoff is matrix as shown in table 1 and P is the probability estimate of the state S_j

Example 3: The Manufacturing Choice (Decision under Risk)

Problem Statement: A manufacturing firm must decide between two production processes for a new gadget.

Process A (Automated): High fixed costs, low variable costs. It performs well in high demand but poorly in low demand.

Process B (Manual): Low fixed costs, high variable costs. It is stable but less profitable in high demand.

The market demand (the state of nature) is uncertain. The marketing department estimates a 60% probability of High Demand and a 40%

probability of Low Demand. The estimated profits (payoffs in thousands of rupees) are arranged in the following table 2.

Table 2: Payoff matrix for a manufacturing firm

Alternatives	High Demand (60%)	Low Demand (40%)
Process A	Rs.200	-Rs.50 (Loss)
Process B	Rs.100	Rs.30

Objective: Determine which process the firm should choose based on the Expected Monetary Value (EMV) criterion.

Solution:

We calculate the EMV for each alternative by taking the sum of profits weighted by their probabilities.

1. Calculate EMV for Process A:

$$EMV(A) = 200 * 0.6 - 50 * 0.4 = 120 - 20 = Rs.100/-$$

2. Calculate EMV for Process B:

$$EMV(B) = 100 * 0.6 + 30 * 0.4 = Rs. 72/-$$

3. Decision:

Comparing the results, $EMV(A) = Rs.100$ and $EMV(B) = Rs.72$.

Since $100 > 72$, the firm should choose Process A. Although Process A carries a risk of loss, its high upside in the likely "High Demand" scenario makes it the statistically superior choice over time.

More precisely, decision theory describes how an agent should act. In practice, we try to figure out how humans make decisions. The descriptive theory of decision-making by humans would greatly impact fields like robotics or economic. However, there is experimental evidence indicating that people systematically violates the axioms of utility theory. Psychologists Tversky and Kahneman explained it with well suited example.

Example 4: Suppose a person in this experiment has to make choice between lotteries A and B, and then between C and D as mentioned in the table 3.

Table 3 : Probability and Payoff value of the choices

A:	Wining Amount of Rs.4000/-with 80% chance
B:	Wining Amount of Rs.3000/-with 100% chance

C:	Wining Amount of Rs.4000/- with 20% chance
D:	Wining Amount of Rs.3000/- with 25% chance

The general choice preference should be B over A by majority of the peoples. Similarly choice D is preferred over C as a safe choice which ensures Rs.3000/- with little higher probability and it appears, 5% risk may add the profit of Rs.1000/- . Whereas if we calculate EMV for the given choices, the results would be reverted.

$$EMV(A) = 4000 * 0.8 = 3200$$

$$EMV(B) = 3000 * 1 = 3000$$

Since $EU(A) > EU(B)$, therefore A is preferred over B, which is exactly the reverse

$$EMV(C) = 4000 * 0.20 = 800$$

$$EMV(D) = 3000 * 0.25 = 750$$

The same case is with this case too, the choice C is preferred than D.

One possible interpretation of the above discussion is that human decisions may appear irrational when evaluated strictly against standard utility axioms. In practice, human choices do not always align with a consistent utility function. An alternative explanation is that such analyses often fail to account for regret. Individuals may anticipate the regret associated with foregoing a certain reward (state B) in favor of an option with an 80% chance of success, particularly if that gamble results in a loss. In other words, choosing option A carries a 20% probability of receiving no monetary payoff and experiencing a strong sense of regret, which significantly influences decision-making beyond expected utility considerations.

Check Your Progress 2

1. What are the different types of environmental scenarios in which an agent performs decision-making?

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2. Discuss two real-time applications in which autonomous agents make decisions based on a given payoff matrix.

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3. Using the following information, determine your choice between Action C (certain) or Action G (gamble) based on expected monetary value.

Action C (certain): receive Rs.50 for sure.

Action G (gamble): 80% chance Rs.100 and 20% chance of complete loss.

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8.4 SEQUENTIAL DECISION PROBLEMS

Payoff matrices are well-suited for static, single-stage decisions; however, many real-world problems require a sequence of decisions over time. A decision made today may reduce some uncertainty but simultaneously create new choices in the future. Such problems are known as sequential decision problems, where each action affects both immediate outcomes and subsequent states of the environment. Because these decisions are interdependent, the optimal choice at any stage depends on past actions and anticipated future consequences. A classic example is the game of chess, where every move influence future possibility and the eventual outcome. The central challenge in such settings is to develop a strategy that optimizes a cumulative objective, such as maximizing total reward or minimizing overall cost across all stages.

Sequential decision problems are categorized according to the nature of the environment and the information available to the agent. When the environment is fully observable, rewards are additive, and state transitions satisfy the Markov property, the problem is modeled as a Markov Decision Process (MDP). In contrast, when the environment is partially observable and the agent does not have complete knowledge of the current state, the problem is formulated as a Partially Observable Markov Decision Process (POMDP).

8.4.1 Markov Decision Processes

We can define MDPs as a fundamental framework for modeling sequential decision problems in which the environment is fully observable and the transitions between states are probabilistic. It models sequential decision problems where an agent interacts with an environment over discrete time steps, aiming to maximize cumulative rewards. The core assumption of an MDP is the Markov Property: the future depends only on the current state and action, not on the sequence of events that preceded it.

An MDP is formally defined by a tuple: $\langle S, A, T, R, \gamma \rangle$, where S = State space, A = actions space, T = Transitions, R = Rewards and γ = Discount Factor, defined as below:

- States (S): The state represents the current environmental situation. It encapsulates all the necessary information to make decisions. For example, in a chess game, the state includes the positions of all pieces on the board, and the agent is assumed to have full observability of the current state $s \in S$ at every time step t .

- **Actions (A):** Actions are the choices available to the agent in any given state. Each action leads to a transition from one state to another in the environment. In the chess example, an action would be to move a piece from one square to another.
- **Transitions (T):** The transition model describes how the state changes in response to an action. This is often probabilistic, particularly in environments where uncertainty is a factor. The probability of transitioning from state s to state s' after applying action a is given below:

$$P(s' | s, a) = \text{Probability of transitioning to } s' \text{ given } s \text{ and } a$$

- **Reward (R):** The reward function assigns a numerical value to each state or state-action pair, representing the immediate benefit of being in that state or performing that action. It is represented as $R(s)$ or $R(s, a)$ or $R(s, a, s')$. The objective is to maximize the cumulative reward over time.
- **Discount Factor (γ):** $\gamma \in [0,1)$ is the discount factor that weights for future rewards. A γ value closer to 0 makes the agent myopic (prioritizing immediate rewards), whereas a γ value closer to 1 makes the agent far-sighted (weights future rewards almost as much as immediate ones).

Optimal Policy (π^*)

A policy (represented as π) is a strategy that defines the action that an agent will take in each state. An optimal policy maximizes the expected cumulative rewards over time. The solution to an MDP is finding an optimal policy π^* , which is a function that tells the agent which action to take in every state to maximize the expected total discounted reward.

$$\pi^*: S \rightarrow A$$

The value of a state s under an optimal policy π^* is given by the Optimal Value Function $V^*(s)$

The optimal value is also known as utility value (described in section 8.2). The Markov property ensures that the next state and reward depend only on the current state and action, enabling dynamic programming solutions like value iteration to find the optimal policy π^* . The value function or utility function allows the agent to select the actions based on optimal value function.

$$(8.1) \quad \pi^*(s) = \operatorname{argmax}_{a \in A} \sum_{s'} T(s, a, s') V(s')$$

where $T(s, a, s')$ is identified by the probability $P(s' | s, a)$

The next section describes the algorithm for computing the optimal value, after which the optimal policy can be determined.

8.4.2 Solving Sequential Decision Problems with Value Iteration

The value of a state is computed as the discounted rewards from that point onwards and is given by,

$$V(s) = R(s) + \gamma \max_{a \in A} \sum_{s'} T(s, a, s') V(s') \quad (8.2)$$

Equation (8.2) is also known as the Bellman equation, after Richard Bellman (1957). This equation is the basis of the value iteration algorithm for solving MDPs. If the number of possible states in an environment is n , then n Bellman equations are required, containing n unknowns, with one equation corresponding to each state. Thus, n simultaneous equations are solved for the unknowns. However, the equations are nonlinear due to the presence of the max operator. Therefore, an iterative approach is used for obtaining the value or utility of the state. We start with the arbitrary initial values for the utilities $V_0(s)$, normally initialized with zero. Let $V_i(s)$ represent the utility value of the state s at i^{th} iteration. The iteration step, to update the Bellman equation, is computed as:

$$V_{i+1}(s) = R(s) + \gamma \max_{a \in A} \sum_{s'} T(s, a, s') V_i(s') \quad 8.3$$

The value iteration algorithm

Begin

Step 1: Initialization

Initialize the value function for all states $s \in S$ to arbitrary values (e.g., zero, or the immediate reward $R(s)$).

$$V_0(s) = 0 \text{ for all } s \in S$$

Set a small convergence threshold $\epsilon > 0$ (e.g., 0.0001).

Initialize the change in value $\delta = \infty$.

Set iteration counter $i = 0$.

Step 2: Iterative Update (Loop)

Repeat while $\delta > 0$

Increment the iteration counter: $i = i + 1$.

Reset the maximum change: $\delta = 0$.

For each state $s \in S$:

Compute the optimal value: Find the maximum expected future discounted return achievable from state s , by choosing the best action a .

$$V_i(s) = \max_{a \in A} (R(s) + \gamma \sum_{s'} P(s' | s, a) V_{i-1}(s'))$$

Update the maximum change: $\delta = \max(\delta, |V_{i-1}(s) - V_i(s)|)$

Step 3: Termination

The loop terminates when the maximum change δ between successive iterations is less than the threshold ϵ . At this point, $V_i(s) \approx V^*(s)$.

End

We can apply the value iteration algorithm to a scenario where a robot is navigating in a two room (considered as two states) hallway: Room1 (s_1) and

Room2 (s2). The complete details and optimal value computation is illustrated in the example below.

Example 5: Let the MDPs be defined using five tuples as follows:

A. States space (S): {s1, s2}

B. Action space (A): {a1, a2}

C. Transition Probabilities is shown by matrix

	a1	a2
s1	s2, $P(s2 s1,a1)=1$	s1, $P(s1 s1,a2)=1$
s2	s1, $P(s1 s2,a1)=1$	s2, $P(s2 s2,a2)=1$

D. Reward is shown by reward matrix

	a1	a2
s1	10	2
s2	3	8

E. Discount Factor (γ): 0.9

Iteration 0:

$$V_0(s1) = 0$$

$$V_0(s2) = 0$$

Iteration 1:

$$V_1(s1) = \max (R(s1, a1) + \gamma P(s2|s1,a1) V_0(s2), R(s1, a2) + \gamma P(s1|s1,a2) V_0(s1))$$

$$= \max (10 + 0.9 (1 \times 0), 2 + 0.9 (1 \times 0))$$

$$= \max (10, 2) = 10$$

$$V_1(s2) = \max (R(s2, a1) + \gamma P(s1|s2,a1) V_0(s1), R(s2, a2) + \gamma P(s2|s2,a2) V_0(s2))$$

$$= \max (5 + 0.9 (1 \times 0), 8 + 0.9 (1 \times 0))$$

$$= \max (5, 8) = 8$$

Iteration 2:

$$V_2(s1) = \max (R(s1, a1) + \gamma P(s2|s1,a1) V_1(s2), R(s1, a2) + \gamma P(s1|s1,a2) V_1(s1))$$

$$= \max (10 + 0.9 (1 \times 8), 2 + 0.9 (1 \times 10))$$

$$= \max (17.2, 11) = 17.2$$

$$V_2(s_2) = \max (R(s_2, a_1) + \gamma P(s_1|s_2, a_1) V_1(s_1), R(s_2, a_2) + \gamma P(s_2|s_2, a_2) V_1(s_2))$$

$$= \max (5 + 0.9 (1 \times 10), 8 + 0.9 (1 \times 8))$$

$$= \max (14, 15.2) = 15.2$$

This process continues until the values converge to the threshold value of ϵ . Once converged, the optimal policy can be extracted. For example, if $V^*(s_1)$ and $V^*(s_2)$ are the final converged values, then for s_1 , the optimal action would be a_1 because it yields a higher expected value.

Partially Observable Markov Decision Processes (POMDPs)

In many real-world scenarios, agents do not have complete information about the current state of the environment. POMDPs extend MDPs by introducing hidden states and an observation model, thereby complicating the decision-making process.

Applications of Sequential Decision Problems:

- **Robotics:** Tasks such as navigation, path planning, and manipulation involve sequential decision problems. To accomplish a task, like arriving at a destination or putting together a product, robots must make a series of decisions while taking dynamic changes in the environment into account.
- **Finance:** Financial decision-making frequently involves sequential choices, such as portfolio management, where investors must choose how to distribute assets over time in order to maximize returns while controlling risks.
- **Healthcare:** Choosing a set of therapies that maximize patient outcomes over time can be modelled as a sequential decision problem in the context of healthcare treatment planning for chronic diseases.
- **Autonomous Systems:** To navigate challenging situations, avoid obstacles, and accomplish their goals, autonomous cars, drones, and other autonomous systems rely on sequential decision-making.

Challenges in Solving Sequential Decision Problems

- **Computational Complexity:** The computational complexity of determining the best course of action increases exponentially with the number of states and actions. We call this the "curse of dimensionality."
- **Uncertainty and Exploration:** In many sequential decision problems, the agent must deal with uncertainty regarding the environment and the outcomes of its actions. One of the main challenges in this area is striking a balance between exploitation (using information to make decisions) and exploration (collecting information).
- **Scalability:** For large-scale issues, traditional approaches might not be practical. High-dimensional state and action spaces are frequently handled using approximation methods like deep reinforcement learning.

Check Your Progress 3

1. What is a sequential decision problem? How does it differ from a single-step decision problem?

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2. Explain how sequential decision problems are represented in AI.

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3. An AI agent is in state S_1 and can take one of two actions: A_1 or A_2 .

If A_1 is taken:

Moves to state S_2 with probability 0.7 and reward = 10

Moves to state S_3 with probability 0.3 and reward = 0

If A_2 is taken:

Moves to state S_2 with probability 0.4 and reward = 6

Moves to state S_3 with probability 0.6 and reward = 4

Assume S_2 and S_3 are terminal states.

Which action should the agent choose based on expected utility?

.....

.....

.....

8.5 ELEMENTARY GAME THEORY

In the previous section, you have gone through the concept of utility and the decision-making process. We have discussed the problems whose solutions are constructed through appropriate decisions that are taken in a sequential fashion. This section explains the foundations of game theory relevant to AI, shows how these ideas are used in multi-agent systems, where multiple decisions are made through interactions among different entities, and illustrates the concepts with examples.

8.5.1 Introduction to game theory

Game theory is a mathematical framework for modeling situations in which multiple decision-makers (called agents or players) interact, and the outcome for each agent depends not only on their own actions but also on the actions of others. In Artificial Intelligence (AI), game theory plays a crucial role in designing intelligent agents that can reason strategically about other agents, whether they are cooperative, competitive, or partially aligned.

In traditional AI, many problems are solved as single-agent optimization tasks, such as finding the shortest path, classifying images, or predicting future values. However, modern AI systems increasingly operate in multi-agent environments: autonomous vehicles interacting on roads, trading agents in

financial markets, robots cooperating in warehouses, agents bidding in online auctions, and security systems defending against attackers. In these environments, an agent's best action depends on what others are likely to do. Game theory provides the language and tools to model these interactions, predict outcomes, and design algorithms that learn equilibrium strategies.

8.5.2 Key Concepts in Game Theory

A game is formally defined using several key components, primarily players, actions, and payoffs.

A game consists of:

- a. Players (agents):** The decision-makers (e.g., robots, software agents, humans).
- b. Actions (or strategies):** The possible choices available to each player.
- c. Payoffs (utilities or rewards):** Numerical values that represent how good an outcome is for each player.

In AI, payoffs are often defined in terms of rewards in reinforcement learning, task completion, cost reduction, or accuracy. Agents aim to choose strategies (or policies) that maximize their expected payoff. Games may be classified based on their structure and the nature of players' interactions. Some important classifications of games are:

- **Cooperative vs. Non-cooperative:** In cooperative games, binding agreements and coalitions are allowed. In non-cooperative games, each player acts independently.
- **Zero-sum vs. General-sum:** In zero-sum games, one player's gain is exactly the other's loss. In general-sum games, players' interests may be aligned, misaligned, or mixed.
- **Simultaneous vs. Sequential:** In simultaneous-move games, players choose actions at the same time without knowing others' moves. In sequential games, players move in turns, often with some knowledge of previous actions.
- **One-shot vs. Repeated:** One-shot games are played once. In repeated games, the same interaction occurs many times, allowing for learning, reputation, and more complex strategies.

A game representation defines who the agents are, what they can do, what they know, and how outcomes are evaluated. Different representations are used to evaluate strategies and determine the final outcome.

Normal-form Representation

A normal-form game (also called strategic-form) is typically represented using a payoff matrix. For two players, Player 1's strategies are along the rows and Player 2's strategies are along the columns. Each cell contains a pair of numbers: (payoff to Player 1, payoff to Player 2).

Table 4: Example of a Normal-form Payoff Matrix

Player 1 / Player 2	C1	C2
R1	(2, 3)	(0, 0)
R2	(1, 1)	(3, 2)

In Table 4, Player 1 chooses between R1 and R2, while Player 2 chooses between C1 and C2. The entry (2, 3) indicates that if Player 1 chooses R1 and Player 2 chooses C1, then Player 1's payoff is 2 and Player 2's payoff is 3.

Extensive-form Representation

Sequential games are often represented using game trees. The nodes represent decision points, edges represent actions, and leaves represent final outcomes and payoffs. AI agents can use tree search algorithms (similar to minimax in adversarial search) to compute optimal strategies in such trees.

Root: Player 1 chooses between strategy A and strategy B.

– If strategy A: Player 2 chooses between X and Y.

* (A, X) → Payoffs (3, 2)

* (A, Y) → Payoffs (1, 4)

– If strategy B: Game ends with payoffs (3, 2)

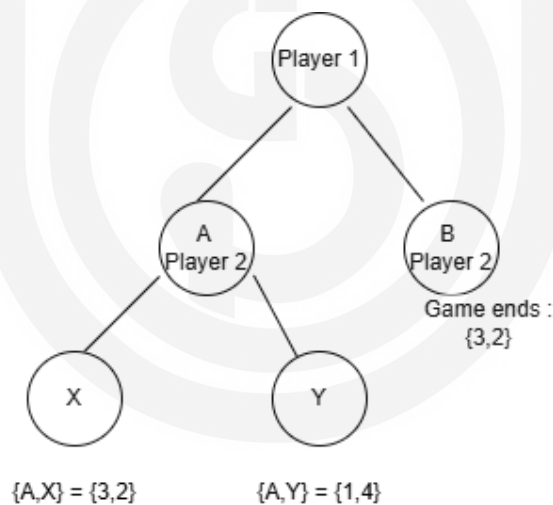


Figure 1. Sample game tree representation

8.5.3 Nash Equilibrium

A Nash equilibrium is a strategy profile (one strategy for each player) such that no player can increase their payoff by unilaterally deviating from their current strategy, assuming the others keep their strategies fixed.

Formally, if $s^* = (s_1^*, s_2^*, \dots, s_n^*)$ is a Nash equilibrium for players 1 to n , then for every player i and any alternative strategy s_i :

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$$

where s_{-i}^* denotes the strategies of all players except i , and u_i denotes player i 's payoff.

Nash equilibria can be in pure strategies (choosing one action with probability 1) or in mixed strategies (randomizing over actions). In some games, no pure strategy Nash equilibrium exists. Agents may then randomize over actions. A **mixed strategy** assigns a probability distribution over the pure actions. The equilibrium condition requires that each player's mixed strategy makes the other players indifferent between the actions they randomize over. In AI, game-theoretic reasoning is used to design multi-agent reinforcement learning algorithms where each agent learns a policy in the presence of other learning agents. Game theory also models adversarial environments, such as security games (defender vs attacker), and adversarial training of models. It creates mechanisms and auctions where self-interested agents interact. Let's discuss an example of a coordination game in AI traffic arrangement.

Example 6: Two-Agent Coordination Game (Traffic AI)

Consider two autonomous vehicles, Car A and Car B, approaching an intersection without traffic lights. Each car can choose between the following two actions:

- a. Go (G)
- b. Wait (W)

The payoffs are defined as follows:

- If both cars Go (G, G), they risk collision: payoff = $(-10, -10)$.
- If one Goes and the other Waits: the one who Goes gets 5 (passes quickly), the one who Waits gets 3 (safe but delayed).
- If both Wait (W, W), both are safe but slightly delayed: payoff = $(2, 2)$.

Table 5: Payoff Matrix for Example 6

Car A / Car B	G	W
G	$(-10, -10)$	$(5, 3)$
W	$(3, 5)$	$(2, 2)$

Step 1: Best Responses

- If Car B chooses G, Car A's payoffs are: $G \rightarrow -10$, $W \rightarrow 3$. So Car A prefers W.
- If Car B chooses W, Car A's payoffs are: $G \rightarrow 5$, $W \rightarrow 2$. So Car A prefers G.

Similarly, for Car B:

- If Car A chooses G, Car B's payoffs are: $G \rightarrow -10$, $W \rightarrow 3$. So Car B prefers W.
- If Car A chooses W, Car B's payoffs are: $G \rightarrow 5$, $W \rightarrow 2$. So Car B prefers G.

Step 2: Nash Equilibria

We look for action pairs where each player is playing a best response to the other:

- (G, G): Both can switch to W for a better payoff, so not an equilibrium.
- (G, W): Car A's best response to W is G (correct), and Car B's best response to G is W (correct). So (G, W) is a Nash equilibrium.
- (W, G): Symmetric reasoning shows (W, G) is also a Nash equilibrium.
- (W, W): Each player could deviate to G and gain more (3 instead of 2). So (W, W) is not an equilibrium.

This game has two pure Nash equilibria: (G, W) and (W, G). They represent asymmetric coordination: one car goes, the other waits. In AI, a coordination mechanism (such as communication or conventions) is needed to avoid collision and to select one of these equilibria consistently.

Check Your Progress 4

1. What is game theory in Artificial Intelligence? Explain its main components.

2. Distinguish between normal form and extensive form representations of a game.

3. Define Nash Equilibrium and explain its importance in AI.

8.6 SAMPLE APPLICATIONS

Utility theory, sequential decision making, and game theory provides AI with a rich set of tools to model strategic interactions among multiple agents. By understanding concepts such as Nash equilibrium, mixed strategies, sequential games, and repeated games, AI practitioners can design agents that behave intelligently in competitive, cooperative, and adversarial environments. Some applications illustrate how AI agents can reason about others, anticipate their responses, and adapt their own strategies.

Self-Driving Cars: The utility function in a self-driving car application may assign utility values to various routes depending on variables like time taken, fuel consumption, safety, and comfort. The self-driving car can then select the route with the highest expected utility to get to its destination by using these utility values to evaluate each route, accounting for the likelihood of various traffic situations or road barriers.

Recommendation Systems: Think of a system that makes movie recommendations for people based on their tastes. Movies that match a user's preferred genres, actors, or directors may receive higher utility values from the recommendation system's utility function, while those that don't may receive lower utility values. The user's utility values can then be used by the recommendation system to rank and suggest films to them, with higher utility films being suggested more prominently.

Multi-agent Reinforcement Learning (MARL): In MARL, multiple agents learn policies simultaneously in a shared environment. The reward of one agent may depend on the actions of others, which creates a game-theoretic structure. Algorithms such as independent Q-learning, Nash Q-learning, and policy gradient methods are often analyzed using game-theoretic tools.

Adversarial and Security Settings: Security games model interactions between a defender (e.g., a security agency or system) and attackers. The defender allocates limited resources to protect targets, while attackers choose targets to attack. AI systems use game-theoretic optimization to compute mixed strategies that are unpredictable and robust.

Auctions and Mechanism Design: Online advertising auctions (such as for display ads or keyword-based ads) can be modeled as games in which bidding agents try to maximize their profit. AI agents use game-theoretic strategies to make bidding decisions based on beliefs about other bidders.

Cooperative Multi-agent Systems: In cooperative scenarios (e.g., swarm robotics or distributed sensor networks), agents share a common objective. Game theory is used to ensure fair reward allocation, encourage truth-telling, and prevent free-riding behavior.

8.7 SUMMARY

In Artificial Intelligence, rational decision-making is modeled using mathematical frameworks that help agents choose actions optimally. Utility theory, sequential decision problems, and game theory together form the foundation for intelligent behavior in both single-agent and multi-agent environments.

Utility theory provides a formal way to represent an agent's preferences using a numerical scale called utility. Each possible outcome of an action is assigned a utility value that reflects how desirable it is to the agent. An intelligent agent uses these values to compare alternatives and select actions that maximize expected utility, especially in uncertain environments. Utility theory enables AI systems to make consistent and rational choices by balancing trade-offs such as risk, reward, and cost, and it is widely used in planning, learning, and decision analysis.

Sequential decision problems deal with situations where decisions are made over time, and each action affects future states and opportunities. Instead of choosing a single action, the agent must determine a policy—a strategy that specifies what action to take in every possible state. These problems are commonly represented using state-transition models, such as Markov Decision Processes (MDPs), which include states, actions, transition probabilities, and reward functions. Sequential decision-making is central to reinforcement learning, where agents learn optimal policies through interaction with the environment by maximizing long-term cumulative rewards.

Game theory in AI extends decision-making to environments involving multiple intelligent agents whose actions influence one another. Each agent is assumed to be rational and seeks to maximize its own utility, often leading to strategic interactions. Game theory provides tools to represent such interactions using payoff functions, strategies, and equilibrium concepts like Nash equilibrium. It is applied to both competitive and cooperative scenarios, including auctions, negotiations, resource allocation, and multi-agent reinforcement learning.

Together, these concepts enable AI systems to reason about preferences, plan actions over time, and act strategically in the presence of other agents. They form a unified framework for modeling rational behavior in complex, dynamic, and interactive environments.

8.8 SOLUTIONS/ANSWERS

Check Your Progress 1

1. Advantages of Utility Function in AI

- a. Enables Rational Agent Behavior: Utility functions allow AI agents to act rationally by selecting actions that maximize expected utility.
- b. Provides a Unified Decision Framework: They offer a common framework for decision-making across planning, learning, and game-theoretic AI systems.
- c. Handles Uncertainty and Risk: When combined with probabilities, utility functions help AI agents make decisions under uncertainty.
- d. Supports Sequential Decision-Making: Utility functions guide long-term planning by evaluating future states, not just immediate rewards.
- e. Useful in Multi-Agent Systems: Utilities help model preferences and incentives in game theory, negotiations, and auctions.

Disadvantages of Utility Function in AI

- a. Difficult to Design: Defining an accurate utility function that truly captures goals and preferences is complex. Poorly defined utility functions can lead to unintended or unsafe agent behavior.
- b. Subjectivity of Utilities: Utility values depend on the designer's assumptions and may not reflect real-world priorities.
- c. Oversimplifies Real-World Goals: Human values, ethics, emotions, and social norms are difficult to encode numerically.
- d. Assumption of Perfect Rationality: Utility-based models assume agents are fully rational, which may not hold in real environments.
- e. Computational Cost: Computing and maximizing utility can be expensive in large or complex state spaces.

2. Step 1: Compute Expected Utility of Invest (I)

$$EU(I) = (0.6 \times U(1,50,000)) + (0.4 \times U(60,000)) = 1,14,000$$

Step 2: Compute Utility of Keep (K)

$$EU(K) = U(1,00,000) = 1,00,000$$

Step 3: Compare Utilities: $EU(I) > EU(K)$

Based on the maximum expected utility criterion, the person should choose Invest (I) because it yields a higher expected utility than keeping the money with certainty.

3. The Maximum Entropy Principle helps in decision-making in AI by guiding an agent to choose a probability distribution that is least biased while still being consistent with the known information. The maximum entropy principle helps make rational decisions under uncertainty by choosing the most unbiased probability distribution that satisfies known constraints, thereby ensuring fair and consistent decision-making.

Check Your Progress 2

1. An **agent's decision-making** depends strongly on the **nature of the environment** in which it operates. Different environmental scenarios influence how the agent perceives, reasons, and chooses actions.
 - a. Fully Observable vs Partially Observable Environments

Fully observable: The agent has complete information about the environment's state at all times (e.g., chess).

Partially observable: The agent has limited or noisy information and must make decisions under uncertainty (e.g., medical diagnosis).
 - b. Deterministic vs Stochastic Environments

Deterministic: The next state is completely determined by the current state and action (e.g., puzzle games).

Stochastic: Outcomes involve randomness; the same action may lead to different results (e.g., stock trading).
 - c. Single-Agent vs Multi-Agent Environments

Single-agent: Only one agent acts in the environment (e.g., route planning).

Multi-agent: Multiple agents interact, cooperate, or compete (e.g., traffic systems).

Autonomous agents often operate in environments where multiple actions lead to different outcomes depending on the behavior of other agents or uncertain conditions. A **payoff matrix** provides a structured way to represent the rewards or costs associated with different combinations of actions, enabling the agent to select the most beneficial strategy. Two important real-world applications where autonomous agents make decisions based on payoff matrices are discussed below.

2. (a) Autonomous Vehicle Interaction at Intersections

In intelligent transportation systems, autonomous vehicles must frequently interact with other vehicles at intersections, merging lanes, or negotiating right-of-way. Each vehicle acts as an autonomous agent that must decide whether to yield, accelerate, or maintain speed. The outcome of these actions depends on the behavior of other vehicles.

A payoff matrix can represent the possible strategies and their associated rewards or penalties, such as safety, travel time, or fuel efficiency. For example, if both vehicles accelerate simultaneously, the payoff may be negative due to the risk of collision. If one vehicle yields while the other proceeds, the system achieves a safe and efficient outcome. By analyzing the payoff matrix, each vehicle selects the strategy that maximizes its expected payoff while maintaining safety. This framework is widely used in autonomous driving algorithms and traffic coordination systems.

(b) Resource Allocation in Wireless Communication Networks

In wireless communication systems, multiple devices or network nodes compete for limited bandwidth or communication channels. Each node acts as an autonomous agent deciding whether to transmit, wait, or adjust transmission power. The success of transmission depends on the actions of other devices in the network.

A payoff matrix models the benefits and costs of these choices. For instance, if two devices transmit simultaneously on the same channel, interference may occur, reducing the payoff for both. If one device transmits while the other waits, the transmitting device receives a high payoff due to successful communication. Autonomous agents use such payoff matrices to determine strategies that balance throughput, interference, and energy consumption. This approach is commonly applied in cognitive radio networks and distributed wireless communication systems.

In both applications, payoff matrices enable autonomous agents to systematically evaluate the consequences of different strategies and select actions that maximize their expected rewards while considering the behavior of other agents in the environment.

3. To decide between **Action C (certain)** and **Action G (gamble)**, we compute the **Expected Monetary Value (EMV)** of each action.

Action C (Certain)-You receive **Rs. 50 for sure**.

Check Your Progress 3

1. A sequential decision problem is a decision-making scenario in which an agent makes a series of decisions over time, and each decision affects the future state of the environment and the options available later. The agent's objective is to choose a policy that maximizes long-term cumulative reward or utility. In contrast, a single-step decision problem involves choosing only one action with immediate consequences and no future dependency. Sequential decision problems require planning and foresight, while single-step problems do not.

2. Sequential decision problems are commonly represented using state-space models, particularly Markov Decision Processes (MDPs). An MDP consists of: States (S), Actions (A), Transition Model (T), Reward Function (R), Policy (π). This representation enables AI agents to plan and learn optimal behavior over time.

3. Expected utility of A_1 :

$$EU(A_1) = (0.7 \times 10) + (0.3 \times 0) = 7$$

Expected utility of A_2 :

$$EU(A_2) = (0.4 \times 6) + (0.6 \times 4) = 2.4 + 2.4 = 4.8$$

Decision: Since $EU(A_1) = 7 > EU(A_2) = 4.8$ the agent should choose action A_1 .

Check Your Progress 4

1. Game theory in Artificial Intelligence is a mathematical framework used to model and analyze situations where multiple rational agents interact and where the outcome for each agent depends on the actions of others. In AI, it helps agents make strategic decisions in competitive or cooperative environments. The main components of game theory are: Players (Agents), Strategies, Payoff (Utility) Function, Rules of the Game, and Outcome.

2. Normal Form Representation: Used for simultaneous-move games, Represented using a payoff matrix, focuses on strategies and payoffs rather than the order of moves, Example: Prisoner's Dilemma.

Extensive Form Representation: Used for sequential games, Represented using a game tree. Shows the order of moves, decision points, and information sets. Example: Chess, bargaining games. Thus, normal form emphasizes strategic choices, while extensive form emphasizes timing and information.

3. A Nash Equilibrium is a strategy profile in which no player can improve their payoff by unilaterally changing their strategy, assuming other players keep their strategies unchanged.